

Multiple Choice (10 marks)

1. Buffer-overflow may remain as a bug in apps if _____ are not done fully.
 - (a) boundary hacks
 - (b) memory checks
 - (c) boundary checks
 - (d) buffer checks
2. The four Primary Security Principles related to messages are
 - (a) Confidentiality, Integrity, Non repudiation and Authentication
 - (b) Confidentiality, Access Control, Integrity, Non repudiation
 - (c) Authentication, Authorization, Availability, Integrity
 - (d) Availability, Authorization, Confidentiality, Integrity
3. Failed sessions allow brute-force attacks on access credentials. This type of attacks are done in which layer of the OSI model?
 - (a) Physical layer
 - (b) Data-link Layer
 - (c) Session layer
 - (d) Presentation layer
4. Why apps developed in languages like C, C++ is prone to Buffer-overflow?
 - (a) No string boundary checks in predefined functions
 - (b) No storage check in the external memory
 - (c) No processing power check
 - (d) No database check
5. What tool can be used to perform SNMP enumeration?
 - (a) DNSlookup
 - (b) Whois
 - (c) Nslookup
 - (d) IP Network Browser

True / False (10 marks)

Answer True or False

6. True or False: The OTP key can be computed using the formula $k = m \text{ xor } c$, where m is the message and c is the ciphertext.
7. True or False: Unauthorized network access is not considered a vulnerability specific to the transport layer.
8. True or False: SQL queries based on user input are particularly vulnerable to injection attacks due to improper input validation.
9. True or False: The Firefox browser provides built-in access to the Tor network without any additional tools.
10. True or False: Snort is primarily designed for network analysis in multiprotocol diverse networks.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find nth bell number.
12. Write a function to check the occurrences of records which occur similar times in the given tuples.

End of Paper